

## **HLM Assignment 1: Due April 16 at Noon**

You are encouraged to work in groups of up to three people on this assignment. Please note all group members on the front page of your submission, and have only one of the members submit a file on Canvas. The file must be submitted in MS Word format (Word is available to all students for free through the University). You must use the HLM software for hierarchical modeling.

Your submission should include selected output where relevant along with your discussion of findings. “Selected output” means output from the HLM software that are relevant to the findings you are addressing. That is, be thoughtful about what you present to us, rather than copying and pasting output of every model that you run. Think of this as submitting a working report to a boss or client. You want them to understand what you did and found, but not waste their time. It should be easy for the reader to understand what you are referring to.

If you want to ask questions about the assignment, attend group office hours with Dr. Jean at 4 p.m. on Tuesday April 14. Do not schedule an individual meeting.

### **Part A: Two-level Cross-sectional Modeling**

We are interested in the “achievement gap” between white and minority students in mathematics achievement in 1980. We use the indicator  $\text{Minority} = 1$  if the student is either African American or Hispanic; 0 if not. (The overwhelming majority of the minority students in this 1980 date set were in fact African American and the few Hispanic students in the sample scored similarly to the African American students; the two groups were therefore combined for simplicity.)

The data for this assignment are from the High School and Beyond (HSB) study and available on Canvas. The level-1 data are in `hsb1.sav`, level-2 in `hsb2.sav`. In the HSB data, the variable `SECTOR` indicates whether a school is public or Catholic, coded 1 = Catholic.

1. Obtain a point estimate and standard error for the gap using ordinary least squares regression [you can use the HLM software or any stat package for this part]. Call this the “national gap.”
2. Now estimate the average difference (and standard error) between a minority and a majority student that attend the same school using HLM. Call this the average “within-school gap.” (Hint: centering of the minority indicator correctly is essential to get the right estimate.)
3. Compare the size of the two gaps estimated in (1) and (2). Provide a substantive explanation for why they are different.

4. Consider the extent to which the minority-majority gaps vary from school to school.
  - Provide statistical evidence on whether or not they vary.
  - Do schools with higher achievement tend to have higher or lower minority-majority gaps? Provide statistical evidence (you do not need a hypothesis test).
  - Provide a graph of the empirical Bayes estimates of the gaps, rank ordered in size and color coded by sector. Is there visual evidence that the size of the gaps varies by sector? Explain.
5. Add school sector to your model at level 2 for both the intercept and the minority gap coefficient. Provide an estimate of the average minority gap in public schools and in Catholic schools. Provide a hypothesis test that these gaps are the same.
6. It is reasonable to suggest that the minority gaps might depend on the racial makeup of the school. The indicator variable HIMINTY is coded 1 if minority enrollment in a school exceeds 40%. Add this indicator to your model along with sector for both level 1 terms.
  - Interpret the coefficients for the new HIMINTY terms reported by your model.
  - How does adding racial makeup to the model change our understanding of how school sector affects math achievement? Explain.
7. Write a summary paragraph explaining what you have learned about the minority-majority math achievement gap in the US in 1980.

## Part B. Two-level Growth modeling

You have been hired to study vocabulary growth during the second year of life, and provided with the Huttenlocher study data. The level-1 data are in vocab1hw.dta, level-2 in vocab2hw.dta [Note: these are Stata files and not the same as the ones we used in class. Select the appropriate option when building your MDM file]. Your employers are particularly interested in vocabulary growth around 18 months, when a proposed early learning program would begin.

1. Write down a level-1 model that includes a linear and quadratic growth terms. For this model, age should be centered around 18 months. Define each  $\pi$  and  $e_{ij}$ .
2. Write down a level-2 model in which each level 1 term is allowed to vary randomly. Define each  $\beta$  and each  $r$ .
3. Run the model using Full Maximum Likelihood estimation. Explain in substantive terms what your estimates for each beta means for the shape of your estimated growth trajectories.

4. Add the measure of words spoken by the mother at the 16-month observation (MOMSPEAK) to the model at level 2 for each level 1 term. Define  $\beta_{00}$ ,  $\beta_{01}$ ,  $\beta_{10}$ , and  $\beta_{11}$ .
5. Run the model with MOMSPEAK and explain in a paragraph what you found.
6. Create a model graph to visualize the results of the final model, showing the model-based trajectories for children whose mothers are at the 25<sup>th</sup> and 75<sup>th</sup> percentiles of MOMSPEAK.